EXPERIMENT-2

Markov Decision Process (MDP)

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Course Name: REINFORCEMENT LEARNING

Slot: B

Aim:

To calculate the **Expected Immediate Reward (EIR)** for different state-action pairs in a **Markov Decision Process (MDP)** using transition probabilities and rewards.

Procedure:

1. Define the states **S1, S2, S3** and actions **A1, A2**.
2. Assign the transition probabilities and corresponding rewards for each state-action pair.
3. Calculate the Expected Immediate Reward using the probability and reward values.
4. Repeat the calculation for all state-action pairs.
5. Prepare a summary table containing the **state, action, transition, and expected reward**.
6. Represent the calculated results using a suitable **visualization**.

Python code:

```
# Expected Immediate Reward Calculation

library(grid)

# Calculations

reward <- c(

  0.6*5 + 0.2*(-1),

  0.4*10 + 0.8*(-5),
```

```
  0.7*3 + 0.5*2,
  0.3*7 + 0.5*1,
  0.9*4 + 0.4*0,
  0.1*6 + 0.6*(-2)
)
# Summary table
result <- data.frame(
  State = rep(c("S1","S2","S3"), each=2),
  Action = rep(c("A1","A2"), 3),
  Expected_Reward = reward
)
print(result, row.names=FALSE)
# Visualization data
data <- data.frame(
  State = rep(c("S1","S2","S3"), each=2),
  Action = rep(c("A1","A2"), 3),
  Transition = c(
    "S1 → S2","S1 → S3",
    "S2 → S1","S2 → S3",
    "S3 → S1","S3 → S2"
  ),
  Expected_Reward = reward
)
# Draw visualization
```

```
grid.newpage()

grid.text(

  "MARKOV DECISION PROCESS (MDP)\nExpected Immediate Reward Summary",

  y=0.95,

  gp=gpar(fontsize=18, fontface="bold")

)

x <- 0.08

y <- 0.83

w <- 0.22

h <- 0.09

headers <- c(

  "Current State",

  "Action",

  "Transition",

  "Expected Reward"

)

header_col <- c(

  "steelblue4",

  "darkgreen",

  "darkorange3",

  "purple4"

)

cell_col <- c(

  "lightcyan",
```

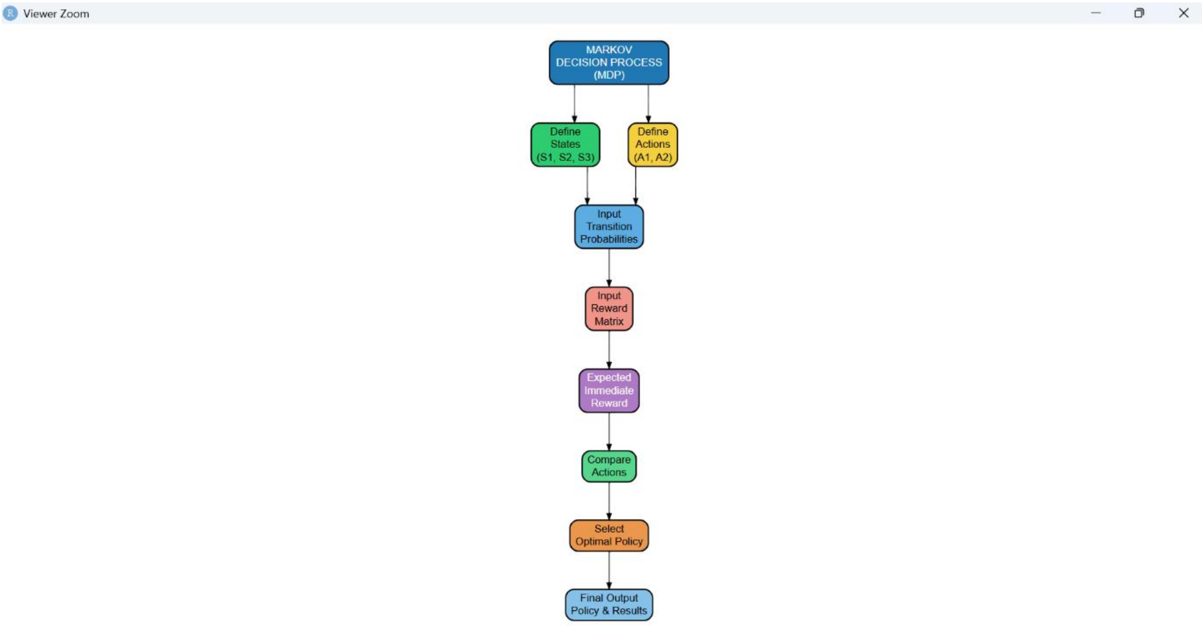
```
  "honeydew",
  "moccasin",
  "lavender"
)
# Headers
for(i in 1:4) {
  grid.rect(
    x=x+(i-1)*w, y=y,
    width=w, height=h, just="left",
    gp=gpar(fill=header_col[i], col="white", lwd=2)
  )
  grid.text(
    headers[i],
    x=x+(i-0.5)*w, y=y,
    gp=gpar(col="white", fontsize=12, fontface="bold")
  )
}
# Table rows
for(r in 1:nrow(data)) {

  yy <- y-r*h
  values <- c(
    data$State[r],
    data$Action[r],
```

```
    data$Transition[r],
    data$Expected_Reward[r]
  )
  for(i in 1:4) {
    grid.rect(
      x=x+(i-1)*w, y=yy,
      width=w, height=h, just="left",
      gp=gpar(fill=cell_col[i], col="black")
    )
    grid.text(
      values[i],
      x=x+(i-0.5)*w, y=yy,
      gp=gpar(
        fontsize=12,
        fontface=ifelse(i==1 || i==2 || i==4,"bold","plain"),
        col=ifelse(i==4,"blue","black")
      )
    )
  }
}
```

Output:

Visualization:



Result:

The **Expected Immediate Reward** was successfully calculated for all state-action pairs. The results were presented in a summary table and visualized to clearly compare the rewards for different actions.